



Pelita Bangsa
Academy

Smart Contract Development on Lisk

BUILDing with IDRX Stablecoin on Lisk Bootcamp

Hi, I'm William from Indonesia



**Institut Sains dan Teknologi
Terpadu Surabaya**

S.Kom. | Information Technology | GPA 4.00



**Alumni Blockchain Developer
Bootcamp Cohort 2**

Web3 Project

(DeFi, DAO, RWA, and many more)

- Pinjoc | CLOB Fixed Rate Lending Protocol
- Talent-chain | Decentralized Service Marketplace
- Wova | Wallet Over Whatsapp

5+ years

Experience in the IT industry

Teaching Experience



Working Experience

(Startup, Corporate, Freelance | Indonesia, Jakarta, Malaysia)



Pelita Bangsa Academy

**Mall of Indonesia, Lantai 2, Unit 2F-A7B,
Jalan Boulevard Barat Raya No. 1,
Kelapa Gading Jakarta Utara 14240.**



”

Upskill dan Reskill Talenta Teknologi Indonesia di bidang Blockchain sehingga kalian bisa bersaing secara global.



SESI

01

**Introduction to Lisk Blockchain
& IDRX Stablecoin**

HARI INI

02

Smart Contract Development on Lisk

26 Mar

03

Interacting with IDRX Stablecoin

9 Apr

SESI

04 Building Frontend UI for Web3 DApp

**Extra
Class**

Introduction to Xellar

05 Integrating Xellar Kit with Smart Contract

06 Brainstorming Web3 Project Ideas

16 Apr

17 Apr

23 Apr

30 Apr

Agenda

**Goal of this
Bootcamp**

Materi + Hands On

Q&A

Wrap Up

Foto Bersama

Recommendation



Physical Notebook



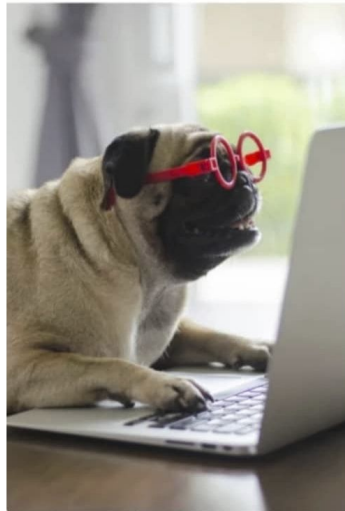
<https://notion.com>

Be Active

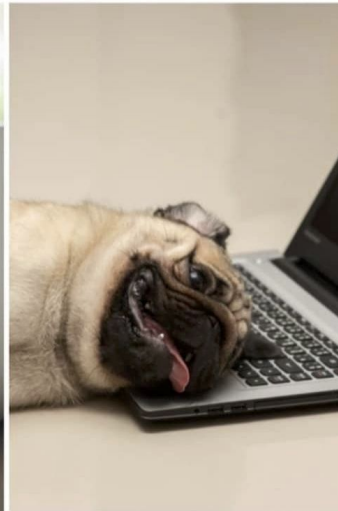


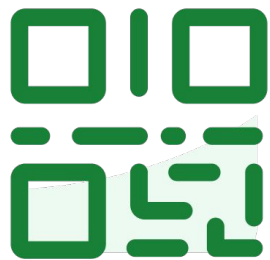
Open Your Camera

Zoom meeting,
with video



Zoom meeting,
audio only





Join at **slido.com**
#235689





Have you ever heard about "Smart Contracts"?





What analogy best describes a Smart Contract?





What excites you most about learning blockchain today?



A

Goal of this Bootcamp



Pelita Bangsa Academy Lisk 1000 IGNITE
BlockDevId X IDR X Xellar

* ♦ *

BUIDLing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

The goal of this bootcamp:

- Learn about Lisk Blockchain Technology
- Come up with Web3 Application Ideas
- #BuildOnLisk Blockchain
- Participate Lisk Builders Challenge Round 1
- Build successful Web3 Startup !!



Fund Tracking Management





Lisk

BlockDevId



IDRX



Xellar



Lisk

Builders
Round 1
Challenge





Project Requirement:

1. Smart Contract deployed on **Lisk Sepolia Testnet** or **Lisk Mainnet** and verified on **Blockscout Explorer**
2. Integrate your dApp with **IDRX Stablecoin** on **Lisk Mainnet** or **Lisk Sepolia Testnet**
3. Build your dApps with **Xellar Kit**
4. Live Project URL

IDRX Address on Lisk Mainnet: 0x18Bc5bcC660cf2B9cE3cd51a404aFe1a0cBD3C22

IDRX Address on Lisk Sepolia Testnet: 0xD63029C1a3dA68b51c67c6D1DeC3DEe50D681661

Prize:

3,000 USD (1st Winner)

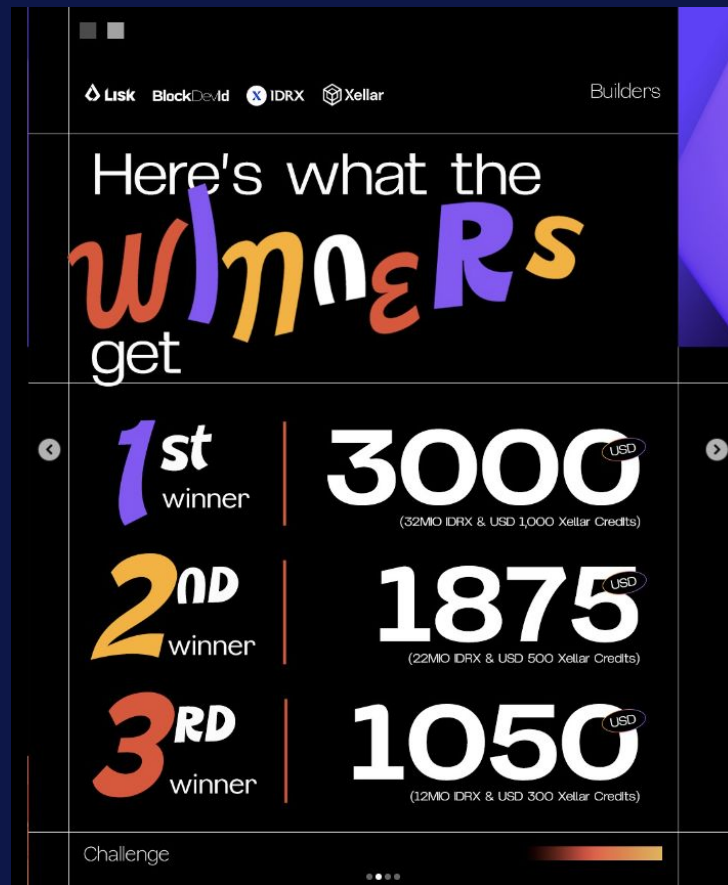
(32 Million IDR^X & USD 1,000 Xellar Credits)

1,875 USD (2nd Winner)

(22 Million IDR^X & USD 500 Xellar Credits)

1,050 (3rd Winner)

(12 Million IDR^X & USD 300 Xellar Credits)

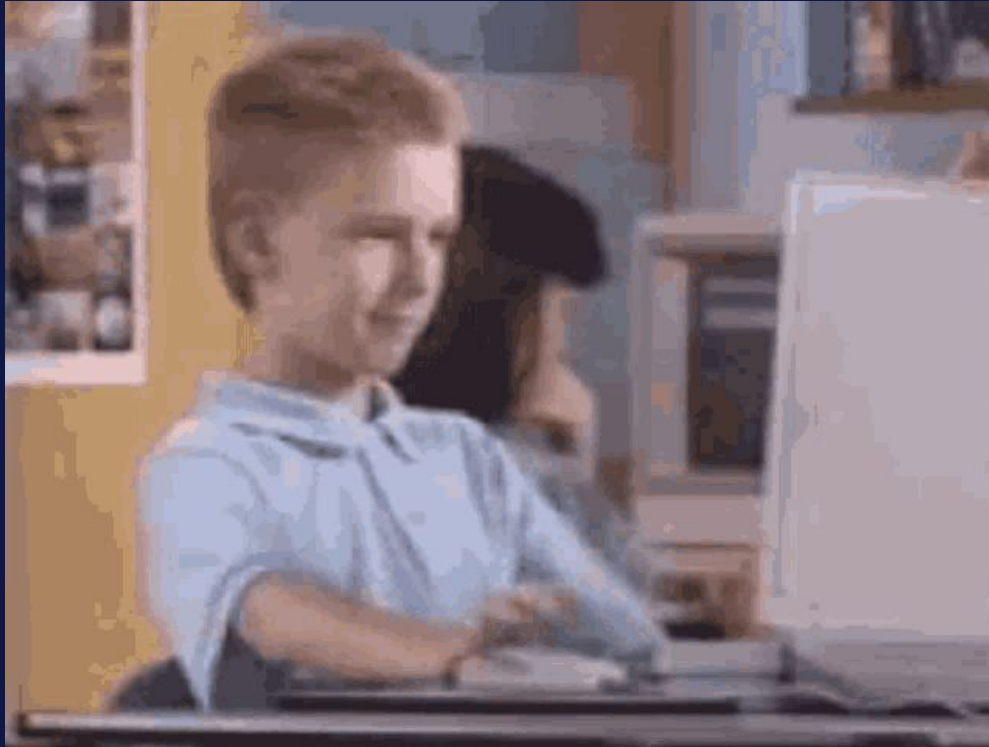


Here's what the winners get

1 st winner	3000 ^{USD} (32MIO IDR ^X & USD 1,000 Xellar Credits)
2 nd winner	1875 ^{USD} (22MIO IDR ^X & USD 500 Xellar Credits)
3 rd winner	1050 ^{USD} (12MIO IDR ^X & USD 300 Xellar Credits)

Challenge

Not bad





Prize:

8 Million IDR^X (Most Favorite Project)

3 Million IDR^X (SocMed Challenge Winner #1)

3 Million IDR^X (SocMed Challenge Winner #2)

The screenshot displays the Lisk Builders challenge interface. At the top, the navigation bar includes the Lisk logo, BlockDevId, IDR^X, Xellar, and a Builders link. The main content area is divided into three sections. The top section features the text 'More CHANCES to win' with 'CHANCES' in large, colorful, stylized letters. The middle section highlights the 'Most Favorite Project' prize, showing '8MIO' (8 Million IDR^X) and '(500 USD)' in a purple font, with a 'Social Media Challenge' banner below it. The bottom section shows the 'Most Engaging & Creative' prize, with '3MIO' (3 Million IDR^X) and '(187.5 USD/each)' in a yellow font, and 'For 2 Winners' below it. The bottom navigation bar includes a 'Challenge' link and a series of dots indicating the current position in the challenge sequence.

Builders

More
CHANCES
to win

Most Favorite Project

8MIO^{IDRX}
(500 USD)

Social Media Challenge

Most Engaging & Creative

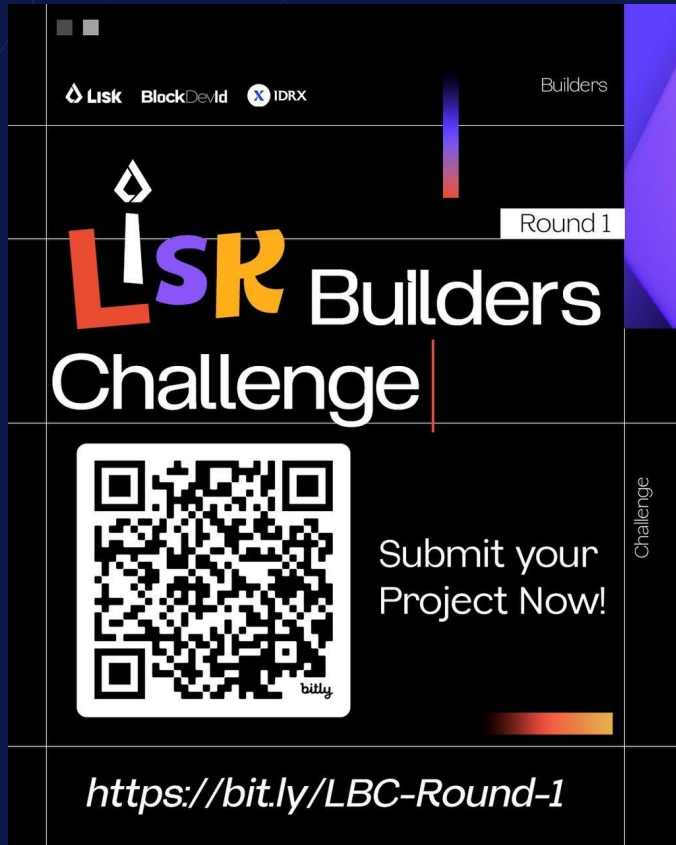
3MIO^{IDRX}
(187.5 USD/each)

For 2 Winners

Challenge

YESSSS!!





The poster is for the Lisk Builders Challenge Round 1. It features the Lisk logo at the top left, followed by 'BlockDevId' and 'IDRX' logos. The word 'Builders' is in the top right. The main title 'Lisk Builders Challenge' is in the center, with 'Round 1' in a small box to the right. Below the title is a QR code and the text 'Submit your Project Now!'. At the bottom is the URL 'https://bit.ly/LBC-Round-1'. The word 'Challenge' is written vertically on the right side of the poster.

Builders

Round 1

Lisk Builders Challenge

Submit your Project Now!

<https://bit.ly/LBC-Round-1>

Challenge

Timeline:

- **Hackathon Start:** March 8, 2025
- **Kickoff AMA:** March 12, 2025
- **Workshop:** March 19 – April 30, 2025
- **Submission Deadline:** May 17, 2025
- **Announcement of Finalists:** May 24, 2025
- **Finalists Coaching:** May 26–30, 2025
- **Demo Day (Jakarta – Hybrid):** May 31, 2025



B

Pelita Bangsa Academy Resources



Pelita Bangsa Academy Lisk IDRX IGNITE
BlackDevid X IDRX Xellar

* ♦ *

BUIDLing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 **GRATIS!**

LIVE via Zoom

→ Swipe untuk daftar

Pelita Bangsa Academy Resources:

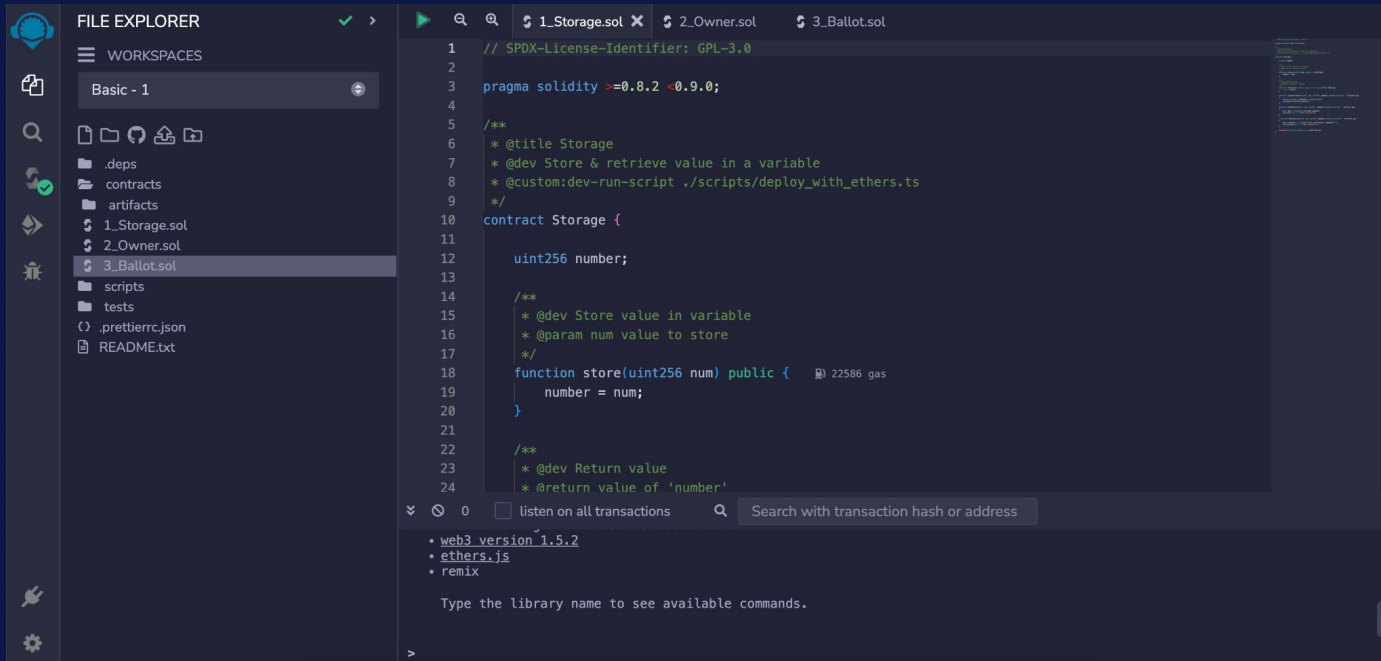
- **LMS (Learning Management System):**
<https://lms.pelitabangsa.co.id/>
(Class Recording, PDF Material & Other Links)
- **Discord Link:**
<https://discord.gg/Jquvw5DqEv>
(Discussion on #lisk channel)

01

Development Environment

Integrated Development Environments (IDE)

Remix



The screenshot displays the Remix IDE interface with a dark theme. On the left, the 'FILE EXPLORER' sidebar shows a project named 'Basic - 1' with a file tree containing folders for '.deps', 'contracts', 'artifacts', and 'scripts', along with files '1_Storage.sol', '2_Owner.sol', '3_Ballot.sol', '.prettierrc.json', and 'README.txt'. The '3_Ballot.sol' file is selected. The main editor area shows the code for '1_Storage.sol' with the following content:

```

1 // SPDX-License-Identifier: GPL-3.0
2
3 pragma solidity >=0.8.2 <0.9.0;
4
5 /**
6  * @title Storage
7  * @dev Store & retrieve value in a variable
8  * @custom:dev-run-script ./scripts/deploy_with_ethers.ts
9  */
10 contract Storage {
11
12     uint256 number;
13
14     /**
15      * @dev Store value in variable
16      * @param num value to store
17      */
18     function store(uint256 num) public { 22586 gas
19         number = num;
20     }
21
22     /**
23      * @dev Return value
24      * @return value of 'number'
25     */
26 }
```

At the bottom, the 'Console' panel shows the installed libraries: 'web3 version 1.5.2', 'ethers.js', and 'remix'. Below the list, it prompts the user to 'Type the library name to see available commands.'.

Remix



SOLIDITY COMPILER

COMPILER +

0.8.22+commit.4fc1097e

☐ Include nightly builds

☒ Auto compile

☐ Hide warnings

Advanced Configurations

Compile 1_Storage.sol

Compile and Run script

CONTRACT

Storage (1_Storage.sol)

Publish on Ipfs

Publish on Swarm

Compilation Details

ABI Bytecode



DEPLOY & RUN TRANSACTIONS

ENVIRONMENT

Remix VM (Shanghai)

VM

ACCOUNT

0xAb8...35cb2 (100 ether)

GAS LIMIT

3000000

VALUE

0 Wei

CONTRACT

Storage - contracts/1_Storage.sol

evm version: shanghai

Deploy

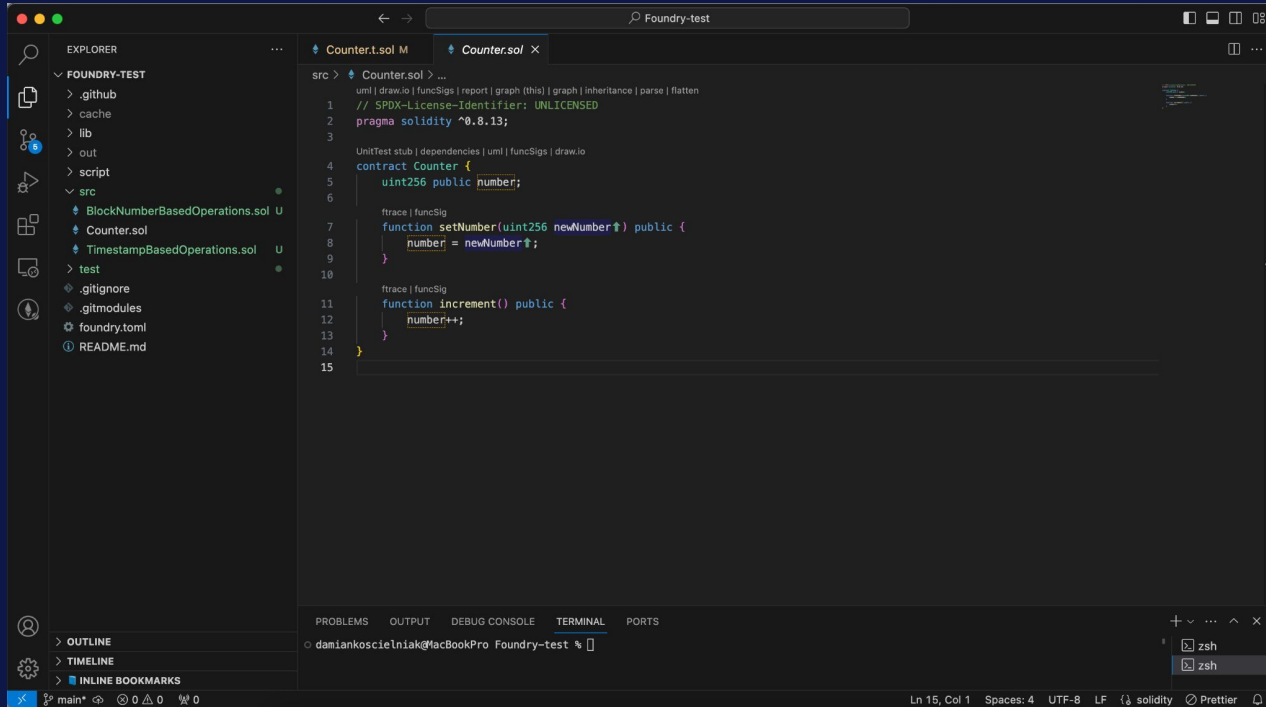
☐ Publish to IPFS

At Address Load contract from Address

Transactions recorded 0

Deployed Contracts

Visual Studio Code



Smart Contract Development Framework

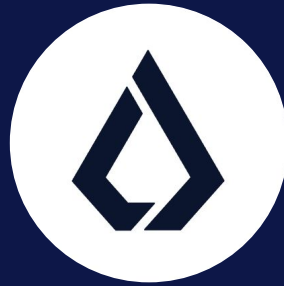


Hardhat



Foundry

EVM Test Networks



Lisk Sepolia

Block Explorers

ETH Price: \$2,080.39 (-1.15%) Gas: 21 Gwei

Search by Address / Txn Hash / Block / Token / Domain Name

Etherscan

HomeBlockchainTokensNFTsResourcesDevelopersMoreSign In

Block #18649117

Overview

Consensus Info

MEV Info

Comments

Block Height:

18649117 < >

Status:

Unfinalized

Timestamp:

10 secs ago (Nov-25-2023 01:49:59 PM +UTC)

Proposed On:

Block proposed on slot 7841348, epoch 245042

Transactions:

146 transactions and 43 contract internal transactions in this block

Withdrawals:

16 withdrawals in this block

Fee Recipient:

beaverbuild in 12 secs

Block Reward:

0.017388803153572673 ETH (0 + 0.306189321686159377 - 0.288800518532586704)

Total Difficulty:

58,750,003,716,598,352,816,469

Size:

98,120 bytes

02

Introduction to Danantiri

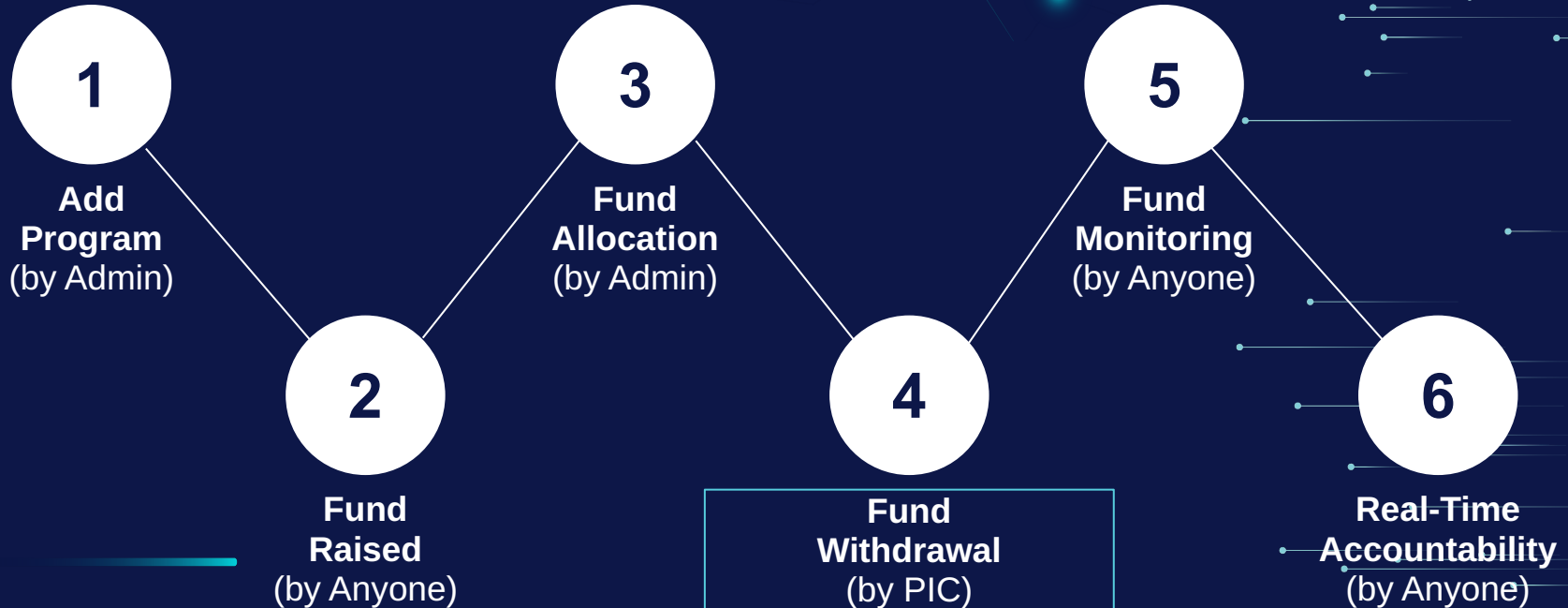
The Problem

- Transparency Issues
- Corruption Risk
- Delayed Reporting

Solution - Danantiri

- Immutable Record
- IDRX stablecoin
- Transparent
- Real-Time Accountability

Business Process



Danantiri | Top Secret



Support Meaningful Projects with Crypto

Your allocated fund will be pooled and distributed to impactful programs.

[Contribute](#)

Managed Fund Amount

12.5 IDR

Programs Funded

4

Featured Programs

MBG

Biar anak sekolah makan enak

0x1RGT7154748V1CE5176479CD0B574AE2448939B5

Testing Program

Testing Program

0x1473289472389578239dfjes

Pembangunan Jalan Tol

Pembangunan Jalan Tol untuk Kontainer

0x3432497DFUOWERN343248098

Pembangunan Sekolah

Pembangunan Sekolah Rakyat

0x3423U43IO24N3B2340934857

Danantiri

Blockchain-Based Government Fund Tracking

Supported Cryptocurrencies

💎 IDRX Lisk Sepolia Testnet

Connect With Us

🗨 Discord PBA

Danantiri | Top Secret

Contribute to Danantiri

Enter the amount you want to contribute.

Enter amount in IDR

Approve IDR

Contribute

Danantiri | Top Secret

[Connect Wallet](#)

Admin Panel - Fund Programs Management

Add New Program

[Add Program](#)

Current Programs

MBG

Biar anak sekolah makan enak

0x1RG77154748V1CE5176479CDG8574AE244B939B5

Testing Program

Testing Program

0x1473289472389678239df/sas

Danantiri | Top Secret

×

MBG

Biar anak sekolah makan enak

PIC Address:
0x1RGT7154748V1CE5176479CDGB574AE244B939B5

Allocate

Remaining Fund:

0 IDRX / 25,000,000,000 IDRX

Cancel



Danantiri | Top Secret

Danantiri | Top Secret

Pembangunan Sekolah

Pembangunan Sekolah Rakyat

PIC Address: 0x3423U43IO24N3B234G934857

Remaining Fund:



100,000,000 IDR / 130,000,000 IDR

Withdraw History

Date	Amount	Note
30000000	25-03-2025	Pembayaran Kontraktor

03

Solidity Contract Development

Key Components

- **Pragma Directive:** Defines compiler version compatibility.
- **Contract Declaration:** Defines the contract's name and functionality.
- **Modifiers:** Control function behavior and access.
- **State Variables + Event:** Permanently store data on the blockchain.
- **Functions:** Executable code units within the contract.



Defining the Contract Structure

Pragma Directive:

Specifies the Solidity version.

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Danantiri {
}
```

Defining the Contract Structure

State Variables:

Define data type and container to save the data.

```
address public owner;  
Program[] public programs;  
uint256 public totalManagedFund;  
uint256 public totalAllocated;  
mapping(uint256 => History[]) public programHistories;
```

Struct

Struct:

Set collection of state variable.

```
struct Program {  
    uint256 id;  
    string name;  
    uint256 target;  
    string desc;  
    address pic;  
    ProgramStatus status;  
    uint256 allocated;  
}
```

Implementing Constructor

Constructor:

Initializes owner and IDRX contract
address.

```
constructor() {  
    owner = msg.sender;  
}
```


Implementing Modifiers

Modifier: Restricts access to certain functions, ensuring only the owner can execute them.

```
modifier onlyAdmin() {  
    require(msg.sender == owner, "Only admin can  
    call this function");  
    _;  
}  
  
modifier onlyPIC(uint256 _programId) {  
    require(msg.sender == programs[_programId].pic,  
    "Not PIC of this program");  
    _;  
}
```

Events in Solidity

- **Events** log information in **transaction logs**, making it more gas-efficient than storing data in contract storage.
- Up to **3 parameters** can be marked as **indexed**, allowing for efficient searching and filtering of events.
- Events can be listened to using web3 libraries like **Web3.js** or **Ethers.js**, which allow applications to catch and react to emitted events.

```
event ProgramCreated(uint256 indexed programId, string name, uint256 target, address pic);

event ProgramUpdated(uint256 indexed programId, string name, string desc, address pic);

event FundSent(address indexed sender, uint256 amount);

event FundAllocated(uint256 indexed programId, uint256 amount);

event FundWithdrawn(uint256 indexed programId, address indexed pic, string history, uint256 amount);
```

Error Handling

01

Require

- Digunakan untuk memvalidasi input dan kondisi tertentu yang mendukung logic dari function

02

Revert

- Sama dengan require, tapi lebih fleksibel penggunaannya. Biasanya di dalam if statement

03

Assert

- Digunakan pada akhir function
- Memvalidasi sesuatu yang mustahil terjadi

Adding Functions for Creating Program

Create Program:

Creates fundraised program that could be allocated by the admin.

```
function createProgram(  
    string calldata _name,  
    uint256 _target,  
    string calldata _desc,  
    address _pic  
) external onlyAdmin  
{  
    ...  
}
```

Adding Functions for Updating Program

Update Program:

Update existing program that is not fully funded yet.

```
function updateProgram(  
    uint256 _programId,  
    string calldata _name,  
    string calldata _desc,  
    address _pic  
) external onlyAdmin  
{  
    ...  
}
```

Adding Function for Retrieving All Program

Get All Program:

Retrieve all existing program.

```
function getAllProgram() external view returns (Program[] memory) {  
    ...  
}
```

IDRX Implementation Adding Interface

IERC20: Widely-used standard contract for creating digital tokens (like IDRX, USDC, etc) on blockchain platforms such as Ethereum or Lisk

Interface:

- Defines the functions your contract can call.
- Ensures communication is correct and secure.

```
interface IERC20 {  
  
    function transferFrom(address sender, address recipient, uint256  
amount) external returns (bool);  
  
    function transfer(address recipient, uint256 amount) external  
returns (bool);  
  
    function balanceOf(address account) external view returns  
(uint256);  
  
}
```

IDRX Implementation Adding State + Constructor

Ensure ERC20 interface is initiated
and saved on the blockchain.

* Need to change the constructor

```
IERC20 public idrxToken;  
  
constructor(address _tokenAddress) {  
    require(_tokenAddress != address(0), "Invalid token address");  
    owner = msg.sender;  
    idrxToken = IERC20(_tokenAddress);  
}
```


IDRX Implementation Adding Function Send Fund

Send Fund:

Allows end-user to send IDRX to this smart contract.

```
function sendFund(uint256 amount) external {  
    ...  
}
```

IDRX Implementation Adding Function Allocate Fund

Allocate Fund:

Allows admin to allocate collected fund from the end-user to the registered program.

```
function allocateFund(uint256 _programId) external onlyAdmin {  
    ...  
}
```

IDRX Implementation Adding Function Withdraw Fund

Withdraw Fund:

Allows **PIC** to withdraw allocated fund by the admin. This function also record how and how much funds are used.

```
function withdrawFund(uint256 _programId, string calldata _history,  
uint256 _amount) external onlyPIC(_programId) {  
    ...  
}
```

IDRX Implementation

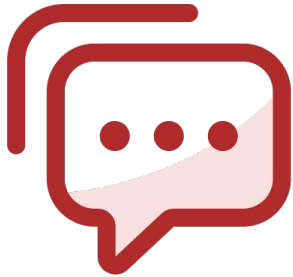
Adding Function Get Program History

Get Program History:

Retrieve all history based on which
parameter program ID sent.

```
function getProgramHistory(uint256 _programId) external view returns  
(History[] memory) {  
    ...  
}
```

Q & A



Audience Q&A

① Click **Present with Slido** or install our
while presenting.

to show live Q&A

Wrap Up

Share 3 things you learn today

On the Zoom Chat

Foto Bersama

Thank You



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Academy**

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